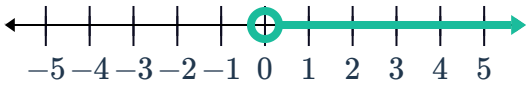
Graphs of linear inequalities

1 State the inequality for x that is represented on the following number lines:

a



b



c



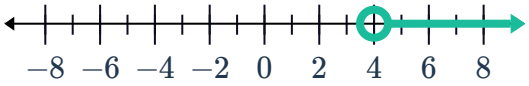
d



e



f



g



h



2 Plot the following inequalities on a number line:

a $x < 0$ **b** $x \geq 6$ **c** $x \leq 1$ **d** $x \geq 0.6$

e $x \geq 0$ **f** $x \leq 2$ **g** $x > 1$ **h** $x < -\frac{3}{4}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Solve inequalities

3 Jenny tried to solve the following inequality but made a mistake in her work:

$$\begin{aligned} \text{Step 0:} \quad -27 - 9x &< 36 \\ \text{Step 1:} \quad -9x &< 63 \\ \text{Step 2:} \quad x &< -7 \end{aligned}$$

Determine which step is incorrect and explain the error.

4 Consider the following inequalities:

i Solve the inequality.

ii Plot the inequality on a number line.

a $7x \leq 63$

b $-4x \leq 20$

c $15 > -3x$

d $x - 1 < 4$

e $1 + x \leq 1$

f $-5 + x \leq 1$

g $6 - x > 10$

h $2x - 8 < 4$

i $-3x + 7 \geq 4$

j $10 - 8x < 26$

k $2(x + 5) > 4$

l $4(2x + 4) > 40$

m $3(4x + 8) \leq -12$

n $\frac{x}{5} > \frac{8}{5}$

o $\frac{x}{6} \geq 2$

p $\frac{x}{3} - 1 \leq -5$

q $2 - \frac{x}{4} \geq 0$

r $\frac{x - 6}{3} \leq -2$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647